

Week 7: Student Performance – Data Cleaning & Transformation

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1. Introduction

The Student Performance dataset includes details about learners, their background information, preparation factors, and examination marks. This task focuses on cleaning, transforming, and organizing the data in Power BI so it can be used effectively for analysis and visualization.

2. Objective

- Import the Student Performance dataset and examine its structure.
- Clean and transform the data with Power Query.
- Build calculated fields including Total Score, Average Score, and Performance Grade.
- Separate students into Pass and Fail categories using the selected criteria.
- Convert subject-wise score columns into a suitable format through Unpivot.
- Develop DAX measures for summarizing student performance.
- Prepare the transformed dataset for dashboard creation and analysis.

3. Dataset Description

The dataset contains the following student and examination attributes:

- Student ID
- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Test Preparation Course
- Math Score
- Reading Score

- Writing Score

These fields provide the information required to study academic performance and observe patterns in students' examination results.

4. Data Import and Inspection

The Student Performance file was brought into Power BI through the Get Data option. Transform Data was then selected to open the dataset in Power Query Editor.

The table was checked for its column structure, data types, missing entries, errors, and duplicate records before starting the transformation process.

5. Data Cleaning

The data was prepared by reviewing duplicate rows, missing or unsuitable values, and the data types assigned to each field. Numerical fields were kept in appropriate numeric formats, while categorical fields were checked for consistent values.

Column names were reviewed and standardized wherever necessary so that the final table would be easier to work with.

6. Categorical Data Transformation

The following categorical fields were checked and standardized:

- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Test Preparation Course

This step improved consistency across categorical values and made them ready for further analysis.

7. Calculated Columns

Three new calculated columns were prepared in Power Query to summarize the students' marks.

Total Score

The Total Score represents the combined marks obtained in Math, Reading, and Writing.

Formula:

$$\text{Total Score} = \text{Math Score} + \text{Reading Score} + \text{Writing Score}$$

Average Score

The Average Score was obtained by taking the Total Score and dividing it by the three subjects.

Custom Column

Add a column that is computed from the other columns.

New column name

Total

Custom column formula ⓘ

= [math score]+[reading score]+[writing score]

Available columns

test preparation course
math score
reading score
writing score
Student ID
Total
Average
performance Grade

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

Custom Column

Add a column that is computed from the other columns.

New column name

average

Custom column formula ⓘ

= [Total]/3

Available columns

parental level of education
gender
race/ethnicity
lunch
test preparation course
Student ID
Total

<< Insert

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✓ No syntax errors have been detected.

OK

Cancel

Formula:

Average Score = Total Score / 3

Performance Grade

A Performance Grade was assigned through conditional rules based on the Average Score.

- A – High Performance
- B – Good Performance
- C – Average Performance
- F – Low Performance

8. Pass/Fail Classification

A Pass/Fail field was created by applying the selected passing condition to the students' subject marks.

The students were grouped into the following two categories:

- Pass
- Fail

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Performance Grade

	Column Name	Operator	Value		Output
If	Average	is greater than or...	90	Then	Grade A
Else If	Average	is greater than or...	75	Then	Grade B
Else If	Average	is greater than or...	50	Then	Grade C
Else If	Average	is greater than or...	40	Then	Grade D
Else If	Average	is less than	40	Then	Fail

Add Clause

Else
null

OK Cancel

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Pass/Fail

	Column Name	Operator	Value		Output
If	Average	is greater than	40	Then	Pass

Add Clause

Else
Fail

OK Cancel

The classification was then used as the basis for calculating the overall pass percentage with DAX.

9. Reshaping Subject Data

The Math Score, Reading Score, and Writing Score fields were selected and converted using the Unpivot Columns option in Power Query.

After the transformation, the resulting fields were renamed as:

- Subject
- Score

Student ID	Attribute	Value
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
1	math score	72
1	reading score	72
1	writing score	74
2	math score	69
2	reading score	90
2	writing score	88
3	math score	90
3	reading score	95
3	writing score	93
4	math score	47
4	reading score	57
4	writing score	44
5	math score	76
5	reading score	78
5	writing score	75
6	math score	71
6	reading score	83
6	writing score	78
7	math score	88
7	reading score	95
7	writing score	92

Student ID	Subject	Score
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	math score	72
1	reading score	72
1	writing score	74
2	math score	69
2	reading score	90
2	writing score	88
3	math score	90
3	reading score	95
3	writing score	93
4	math score	47
4	reading score	57
4	writing score	44
5	math score	76
5	reading score	78
5	writing score	75
6	math score	71
6	reading score	83
6	writing score	78
7	math score	88
7	reading score	95
7	writing score	92

This restructuring makes it possible to compare students' performance across subjects using one common Score field.

10. Data Model Preparation

Once the transformations were completed, the table was reviewed and fields that were not required were removed where appropriate. The final structure was checked to make sure it was suitable for analysis.

The completed transformation was loaded back into Power BI by selecting Close & Apply.

11. DAX Measures

The following measures were developed in DAX to summarize student performance:

- Average Math Score
- Average Reading Score
- Average Writing Score

Measures (2)	
Average Math Score	
Average Reading Score	
Average Score	
Average Writing Score	
Overall Pass %	

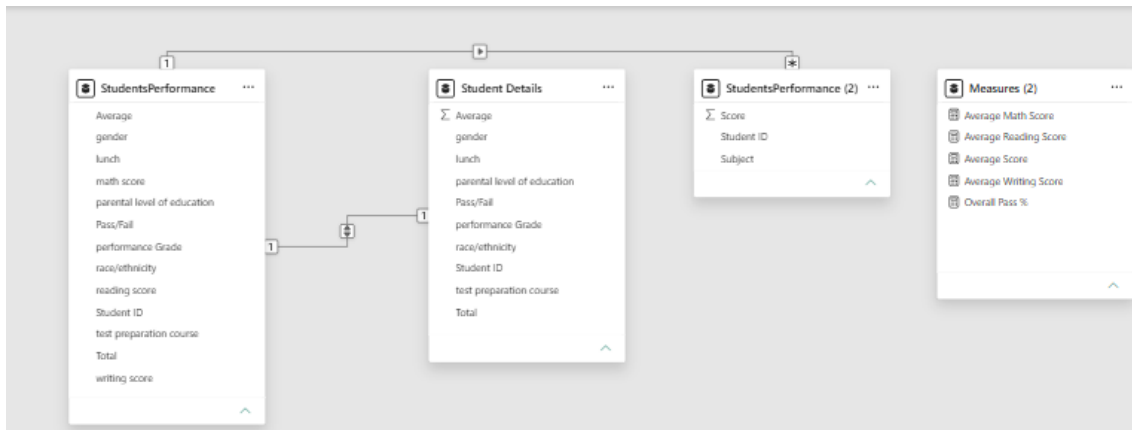
- Overall Average Score
- Overall Pass %

These measures provide concise numerical summaries that can be used for evaluating overall academic performance.

12. Validation

The transformed data was checked against the original dataset to confirm that the calculated values were correct. Total Score and Average Score were reviewed, and the Pass/Fail and Performance Grade categories were checked for consistency.

The unpivoted Subject and Score fields were also inspected to confirm that the subject-wise transformation had been carried out correctly.



13. Conclusion

The Week 7 activity provided practical experience in using Power Query for cleaning and transforming student data and using DAX for analytical calculations. The completed dataset is organized and suitable for creating meaningful visualizations and studying student performance.